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United States Patent Application Publication

20250280913

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Welsh; Geoffrey

METHOD FOR REPLACING A STRAP FOR HELMET SUSPENSION

Abstract

A method is provided for attaching a replacement rear strap to a helmet that includes removing a first rear strap from a headgear assembly which may include a pair of attachment bosses, a top strap, a front strap, and a first rear strap and then attaching the replacement rear strap to the headgear assembly via a pair of adaptor rings configured to be captured on the pair of attachment bosses. The replacement rear strap may consist of stretchable fabric. Each of the adaptor rings may include a adapter ring aperture shaped to fit over the attachment bosses.

Inventors: Welsh; Geoffrey (Lawton, OK)

Applicant: Welshco LLC (Lawton, OK)

Family ID: 1000008626857

Appl. No.: 19/218901

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 18102194 20230127 parent-grant-document US 12329228 child US 19218901

us-provisional-application US 63304752 20220131

Publication Classification

Int. Cl.: A42B3/08 (20060101); A42B3/14 (20060101); A61F9/06 (20060101)

U.S. Cl.:

CPC A42B3/08 (20130101); A42B3/145 (20130101); A61F9/06 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/304,752 filed Jan. 31, 2021, and U.S. Non-Provisional application Ser. No. 18/103,194 titled “Replacement Strap for Helmet Suspension” filed on Jan. 27, 2023. The disclosures of both applications are incorporated by reference into this document as if fully set out herein in their entirety.

FIELD OF THE INVENTION

[0002] This invention relates generally to the field of protective helmets and more particularly, but not by way of limitation, to a method for replacing a strap of a helmet for a helmet suspension.

BACKGROUND

[0003] A wide variety of helmets are used for safety purposes in industrial and recreational activities. Hard hats and welding helmets are commonly worn to protect workers from impact, sparks, and other workplace hazards. For example, FIG. 1 depicts a PRIOR ART welding helmet **200**. The welding helmet **200** includes a face shield **202**, a lens **204**, and a standard headgear assembly **206**. The face shield **202** protects the welder's face from sparks, slag and debris produced during the welding process. The lens **204** is permanently or temporarily tinted to reduce the high intensity light from harming the welder's eyes. The standard headgear assembly **206** supports the helmet **200** on the head of the welder. In some cases, the standard headgear assembly **206** is referred to as a helmet “suspension” because the helmet **200** rides on the standard headgear assembly **206**, which is in turn connected to the welder's head.

[0004] The PRIOR ART standard headgear assembly **206** is depicted in isolation in FIGS. 2 and 3. The standard headgear assembly **206** generally includes a front strap **208**, a standard rear strap **210**, a top strap **212**, and two hubs **214**. In the embodiment depicted in FIGS. 1-3, the front strap **208**, standard rear strap **210** and top strap **212** each extend between and are joined together at the two hubs **214**.

[0005] The front strap **208** may optionally include a cloth sweatband **216** designed to absorb perspiration from the welder's forehead. The standard rear strap **210** includes a rear strap adjustment mechanism **218** that is configured to increase or decrease the length of the standard rear strap **210**, which enables the standard headgear assembly **206** to be tightened around the back of the welder's head. The internal circumference created by the front strap **208**, the standard rear strap **210** and the connecting hubs **214** can be adjusted to match the circumference of the welder's head.

In the embodiment depicted in FIGS. 1-3, the rear strap adjustment mechanism **218** includes a ratcheting system in which a dial is rotated to increase or decrease the length of the standard rear strap **210**. For heavier helmets **200**, it is particularly important for the standard rear strap **210** to be tightened about the welder's head so that the helmet **200** remains securely fastened during use.

[0006] The top strap **212** includes a top strap adjustment mechanism **220**. In the embodiment depicted in FIGS. 1-3, the top strap adjustment mechanism **220** includes a series of adjustment holes and posts that cooperate to adjust the length of the top strap **212**. In this way, the length of the top strap **212** can be adjusted so that the front strap **208** and standard rear strap **210** are correctly positioned on the welder's head.

[0007] The face shield **202** is connected to the standard headgear assembly **206** with fasteners **222**. Each fastener **222** includes a threaded bolt **224** with a bolt base **226**, and a nut **228** that is configured to be hand-tightened onto the threaded bolt **224**. The hubs **214** each include an attachment boss **230** that is designed to contact the inside of the sides of the face shield **202**. Each attachment boss **230** includes a fastener bore **232**, which is configured to receive the threaded bolt **224**. The bolt base **226** is preferably larger than the fastener bore **232**. Each attachment boss **230** also has an outer perimeter defined by side surfaces **234**.

[0008] As best shown in FIGS. 2-3, the attachment bosses **230** extend outward from the hubs **214** such that the bolt base **226** is retained within the interior of the corresponding attachment boss **230** and not in contact with the welder's head. Each of the threaded bolts **224** extends through the corresponding fastener bore **232** and through a mating hole in the face shield **202**, where the nut **228** is secured to the threaded bolt **224** on the outside of the face shield **202**. Tightening the nut **228** onto the threaded bolt **224** applies a compressive force between the attachment boss **230** and face shield **202**.

[0009] In most helmets **200**, the front strap **208**, standard rear strap **210** and top strap **212** are manufactured from a thermoplastic through an injection molding process. In this way, the front strap **208**, standard rear strap **210**, top strap **212**, hubs **214** and attachment bosses **230** can be constructed as a unitary piece that is assembled by connecting two ends of the standard rear strap **210** at the rear strap adjustment mechanism **218** (which may be separately manufactured) and connecting two ends of the top strap **212** at the top strap adjustment mechanism **220**.

[0010] Many helmets **200** are designed such that the face shield **202** can be pivoted from a deployed (lowered) position to protect the welder's face while welding to a retracted (raised) position that permits the welder to see under the raised helmet while preparing for a welding operation. To support the weight of the face shield **202** in the raised position, the nuts **228** must be sufficiently tightened to apply an appropriate amount of compression and friction between attachment bosses **230** and the inside of the face shield **202**.

[0011] Although the basic form and function of the standard headgear assembly **206** depicted in FIGS. 1 and 2 has found widespread acceptance, the standard headgear assembly **206** nonetheless suffers from several deficiencies. In particular, the rear strap adjustment mechanism **218** is prone to failure after extended use. Because the rear strap adjustment mechanism **218** is tightened or loosened each time the helmet **200** is donned or doffed by the welder, the rear strap adjustment mechanism **218** often fails before the balance of the components within the helmet **200**. In some cases, the standard rear strap **210** is uncomfortable, particularly when worn for extended periods. There is, therefore, a need for an improved headgear assembly for use in a hardhat, welding helmet or other helmet that includes a helmet suspension.

SUMMARY OF THE INVENTION

[0012] In one aspect, embodiments of the present disclosure is directed to a method for replacing a rear strap for use on a helmet, where the helmet has a headgear assembly includes a pair of connection bosses. The method includes removing a first rear strap from the headgear assembly and attaching a replacement rear strap assembly to the headgear assembly. The replacement rear strap assembly includes a pair of adapter rings and a replacement strap extending between the pair of adapter rings. In some embodiments, each of the pair of adapter rings includes an adapter ring aperture. In some embodiments, each of the adapter ring apertures is sized and configured to be placed over a corresponding one of the pair of connection bosses. Each of the adapter ring apertures may be D-shaped or circle shaped or other shapes known in the art. In several embodiments, the headgear assembly has a first rear strap that extends between the pair of hubs. In other embodiments, the headgear assembly does not include a standard rear strap that extends between the pair of hubs.

[0013] In another aspect, embodiments of the present disclosure are directed to a method for adding a replacement rear strap assembly to a helmet that includes a standard headgear assembly that has a pair of hubs, a front strap extending between the pair of hubs, and a standard rear strap extending between the pair of hubs. The method includes the steps of removing the standard headgear assembly from the helmet, removing the standard rear strap from the standard headgear assembly to provide a modified headgear assembly, connecting the replacement rear strap assembly to the modified headgear assembly, and connecting the modified headgear assembly and replacement rear strap assembly to the helmet. In some embodiments, the standard headgear assembly further includes a pair of attachment bosses each extending from a corresponding one of the pair of hubs,

and the step of connecting the replacement rear strap assembly includes the step of connecting an adapter ring of the replacement rear strap assembly over each of the pair of attachment bosses. The step of removing the standard rear strap from the standard headgear assembly may further include the step of severing the standard rear strap from the pair of hubs on the standard headgear assembly. In some embodiments, the step of connecting the modified headgear assembly and replacement rear strap assembly to the helmet includes the step of connecting the modified headgear assembly to the helmet with fasteners.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. **1** provides a partial cutaway depiction of a PRIOR ART welding helmet with headgear assembly.

[0015] FIG. **2** is a perspective view of the PRIOR ART headgear assembly of FIG. **1**.

[0016] FIG. **3** is a close-up view of the rear strap and adjustment mechanism from the PRIOR ART headgear assembly of FIG. **2**.

[0017] FIG. **4** depicts a portion of the headgear assembly modified in accordance with an exemplary embodiment of the present invention.

[0018] FIG. **5** depicts an adapter ring and replacement strap of the replacement strap assembly.

[0019] FIG. **6** depicts the replacement strap assembly of FIG. **5** connected to the modified headgear assembly of FIG. **4**.

[0020] FIG. **7** provides a top view of the replacement strap assembly and modified headgear assembly within the helmet.

[0021] FIG. **8** provides rear perspective view of the replacement strap assembly and modified headgear assembly within the helmet.

[0022] FIG. **9** depicts a welder wearing the helmet with the replacement strap assembly.

[0023] FIG. **10** depicts a rear view of the replacement strap assembly in accordance with an exemplary embodiment of the present invention.

[0024] FIG. **11** depicts two adapter rings and a strap clip in accordance with an exemplary embodiment of the present invention.

DETAILED DESCRIPTION

[0025] In accordance with an exemplary embodiment, FIG. **4** depicts a modified headgear assembly **100**. The modified headgear assembly **100** includes the front strap **208**, top strap **212**, hubs **214** and attachment bosses **230** common to the prior art standard headgear assembly **206**, but the standard rear strap **210** and rear strap adjustment mechanism **218** have been removed. In some embodiments, the standard rear strap **210** can be removed simply by severing the standard rear strap **210** from both hubs **214**. Scissors, utility knives or other cutting tools can be used to remove the standard rear strap **210** from the hubs **214**. In other embodiments, a standard headgear assembly **206** may be modified during initial manufacture to omit the standard rear strap **210**.

[0026] FIGS. **5-7** depict a replacement rear strap assembly **102**. The replacement rear strap assembly **102** includes a pair of adapter rings **104** and a replacement strap **106** that extends between the pair of adapter rings **104**. In some embodiments, a 3-D printing process is used to create each adapter ring **104** from any material suitable for a 3-D printer. In a preferred embodiment using a 3-D printing process, each adapter ring **104** is made from polycarbonate. It will be appreciated, however, that each adapter ring **104** may be made by any other suitable manufacturing processes and materials. Further, each adapter ring **104** may be made through the combination of one or more processes and from one or more materials. For example, each adapter ring **104** may be made from any suitable plastic, metal, wood, fabric or textile or any combination thereof.

[0027] Each adapter ring **104** includes an adapter ring aperture **108** that is designed to fit around a

corresponding one of the attachment bosses **230**. As illustrated in FIG. **4**, the attachment boss **230** is “D-shaped” and as illustrated in FIG. **5**, the adapter ring aperture **108** is congruently “D-shaped” and sized to closely fit over the attachment boss **230**, such that the adapter ring aperture **108** corresponds to the outer perimeter of the attachment boss **230**. It will be appreciated, however, that the adapter ring aperture **108** can be configured in any shape to match or accommodate the outer perimeter of a corresponding attachment boss **230**, which may take on a variety of shapes and configurations.

[0028] Each adapter ring **104** further includes an upper surface **105**, a lower surface **107** positioned opposite of the upper surface **105**, and outer side surfaces **109** that extend from edges of the upper surface **105** to corresponding edges of the lower surface **107** and define an outer perimeter of the adapter ring **104**. As shown in FIG. **11**, the upper surface **105** may include a fillet **111** at its edges. In some embodiments, the fillet **111** is between 1 mm to 2 mm in width. In several embodiments, the fillet **111** is 1.2 mm in width. As illustrated in FIGS. **5-6** and FIGS. **10-11**, the outer perimeter of each adapter ring **104** may be a streamlined shape with obtuse angles, curves, and straight edges. Notwithstanding these exemplary embodiments, it will be appreciated that the outer perimeter of each adapter ring **104** can be configured in any shape or size to accommodate the adapter ring aperture **108**. In some embodiments, the outer perimeter may include sharp angles rather than curves. As illustrated in FIG. **11**, each adapter ring **104** may have a shape and size mirroring that of the other. However, it is appreciated that the adapter rings **104** in some embodiments can have a different shape and size from each other.

[0029] Each of the adapter rings **104** can include a strap slit **110** that is sized to permit the replacement strap **106** to be passed through the strap slit **110**. In some embodiments, the strap slit **110** has a height that the same as the width of the replacement strap **106**. In other embodiments, the strap slit **110** has a height that is greater than the width of the replacement strap **106**. In a first preferred embodiment, the replacement strap **106** has a width of 2 inches. In a second preferred embodiment, the replacement strap **106** has a width of 1.5 inches. The replacement strap **106** of FIGS. **7** and **10** may be manufactured to any width and length that is suitable for securely fastening the modified headgear assembly **100** to a user's head during use. The replacement strap **106** has a short end **114** that passes through the strap slit **110** of a first adapter ring **104** and is then attached by stitching or by other suitable attachment devices, such as adhesive, snaps, buttons, clips or other closure mechanisms, to a first position on the replacement strap **106** that is proximate to the first adapter ring **104**. The replacement strap **106** of FIGS. **7** and **10** also has a long end **116** that passes through the strap slit **110** of a second adapter ring **104** and is then attached by stitching or by other suitable attachment devices to a second position on the replacement strap **106** that is distant from the second adapter ring **104**.

[0030] As illustrated in FIGS. **7** and **10**, the replacement strap **106** can include a strap clip **112** that permits the replacement strap **106** to be lengthened or shortened. It is understood, however, that other mechanisms may be used to permit the replacement strap **106** to be lengthened or shortened, such as a hook and loop fastener or a double ring closure. In some of the embodiments, a 3-D printing process is used to create the strap clip **112** from any material suitable for a 3-D printer. In a preferred embodiment using a 3-D printing process, the strap clip **112** is made from polycarbonate. It will be appreciated, however, that the strap clip **112** may be made by any other suitable manufacturing processes and materials and, in fact, may be made through the combination of one or more processes and from one or more materials. For example, the strap clip **112** may be made from any suitable plastic, metal, wood, fabric or textile or any combination thereof. As shown in FIG. **11**, the strap clip **112** may have two strap apertures **113** that are sized to permit the replacement strap **106** to be passed through the strap apertures **113**. In other embodiments, the strap clip **112** may instead have only one strap aperture **113**. In several embodiments, the replacement strap **106** may have multiple strap clips **112**. For example, the replacement strap **106** may have two strap clips **112**, each strap clip **112** having only one strap aperture **113**. The strap clip **112** may also

include a fillet **111** at its edges. In some embodiments, the fillet **111** is between 1 mm to 2 mm in width. In several embodiments, the fillet **111** is 1.2 mm in width.

[0031] In some embodiments, the replacement strap **106** is manufactured from an elastic material that stretches in one or more directions. Suitable strap material is commonly found in the straps used on standard ski goggles and headlamps. In some embodiments, the replacement strap **106** may include a non-slip silicone strip. In several embodiments, an additional reinforcing material, such as leather or rubber, may be used to provide more structure to the replacement strap **106**. The replacement strap **106** may be patterned, colored, or textured to alter the visual appearance.

[0032] As illustrated in FIGS. 7-9, the replacement rear strap assembly **102** is attached to the modified headgear assembly **100** by removing the standard headgear assembly **206** from the helmet **200**, removing the standard rear strap **210** from the standard headgear assembly **206** to provide the modified headgear assembly **100**, securing the replacement rear strap assembly **102** to the modified headgear assembly **100** by placing each of the apter rings **104** over a corresponding one of the attachment bosses **230**, and then attaching the modified headgear assembly **100** to the face shield **202** with the fasteners **222**. In this way, the adapter rings **104** are captured on the attachment bosses **230** between the hubs **214** and the interior of the face shield **202**. The attachment bosses **230** on worn-out or older headgear may develop rough edges that make it difficult to slide hardware over the outer perimeter of the attachment bosses **230**. Therefore, depending on the condition of the standard headgear assembly, it may be necessary to sand or file down these rough edges on the attachment bosses **230** before placing each of the adapter rings **104** over a corresponding one of the attachment bosses **230**. It will be appreciated that the fasteners **222** may be made from any suitable material, including but not limited to metal, plastic, or wood. In most embodiments, the outer side surfaces **109** of the adapter rings **104** have approximately the same width as the side surfaces **234** of the attachment bosses **230**, such that the adapter rings **104** are captured tightly between the hubs **214** and the interior of the face shield **202** during use. In other embodiments, the outer side surfaces **109** of the adapter rings **104** have a smaller width than the side surfaces **234** of the attachment bosses **230**, such that the adapter rings are more loosely captured between the hubs **214** and the interior of the face shield **202**. The replacement rear strap assembly **102** provides an inexpensive, durable, and comfortable modification to the standard rear strap **210** found in conventional headgear assemblies.

[0033] Although the use of the replacement rear strap assembly **102** has been illustrated in connection with the welding helmet **200**, it will be understood that the replacement rear strap assembly **102** can also be used on other hats and helmets, including, but not limited to, hard hats, climbing helmets, ski helmets, bicycle helmets, and motorcycle helmets. Additionally, although the replacement rear strap assembly **102** is designed to replace the standard rear strap **210** of the standard headgear assembly **206**, it will be appreciated that the replacement rear strap assembly **102** can be used in addition to, and without removing, the standard rear strap **210**. In some embodiments, the standard headgear assembly **206** or modified headgear assembly **100** may not include attachment bosses **230**. For those applications, the adapter rings **104** can otherwise be connected to the modified headgear assembly **100** or standard headgear assembly **206** using the fasteners **222** and a smaller adapter ring aperture **108**.

[0034] It is to be understood that even though numerous characteristics and advantages of various embodiments of the present invention have been set forth in the foregoing description, together with details of the structure and functions of various embodiments of the invention, this disclosure is illustrative only, and changes may be made in detail, especially in matters of structure and arrangement of parts within the principles of the present invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed. It will be appreciated by those skilled in the art that the teachings of the present invention can be applied to other systems without departing from the scope and spirit of the present invention.

Claims

1. A method for replacing a first rear strap on a headgear assembly comprising a pair of attachment bosses, the method comprising: removing the first rear strap from the headgear assembly; and attaching a replacement rear strap having a pair of adapter rings to the headgear assembly by engaging each of the adapter rings of the replacement rear strap to an attachment boss of the headgear assembly.
 2. The method of claim 1, wherein the replacement rear strap is made of stretchable fabric.
 3. The method of claim 1, wherein each of the pair of adapter rings comprises an adapter ring aperture.
 4. The method of claim 3, wherein each of the adapter ring apertures is sized and configured to engage an attachment boss.
 5. The method of claim 4, wherein each of the pair of attachment bosses is D-shaped.
 6. The method of claim 5, wherein each of the adapter ring apertures is D-shaped.
 7. The method of claim 4, wherein each of the pair of attachment bosses is circle shaped.
 8. The method of claim 7, wherein each of the adapter ring apertures is circle shaped.
 9. A method for replacing a rear strap of a headgear assembly comprising a top strap, a front strap, a pair of hubs, and a pair of attachment bosses, the method comprising: removing the rear strap from a headgear assembly; and attaching the replacement rear strap having a pair of adapter rings by engaging each of the adapter rings to an attachment boss.
 10. The method of claim 9, wherein the replacement rear strap comprises stretchable fabric.
 11. The method of claim 9, wherein each of the pair of adapter rings comprises an adapter ring aperture.
 12. The method of claim 11, wherein each of the adapter ring apertures is sized and configured to be placed over a corresponding one of the pair of attachment bosses.
 13. The method of claim 12, wherein each of the pair of attachment bosses is D-shaped.
 14. The method of claim 13, wherein each of the adapter ring apertures is D-shaped.
 15. The method of claim 12, wherein each of the pair of attachment bosses is circle shaped.
 16. The method of claim 15, wherein each of the adapter ring apertures is circle shaped.
 17. A method for modifying a helmet comprising a headgear assembly comprising a pair of hubs and a rear strap, the method comprising: removing the headgear assembly from the helmet; modifying the headgear assembly by connecting a replacement rear strap having a pair of adapter rings to the headgear assembly through engagement of each adapter ring with a hub; and connecting the modified headgear assembly to the helmet.
 18. The method of claim 17, wherein the replacement rear strap comprises stretchable fabric.
 19. The method of claim 17, wherein the headgear assembly further comprises a pair of attachment bosses each extending from a corresponding one of the pair of hubs, wherein the step of modifying the headgear assembly further comprises connecting an adapter ring of the replacement rear strap over each of the pair of attachment bosses.
 20. The method of claim 19, wherein the pair of attachment bosses and pair of adapter rings are the same shape.
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